
IAL Chemistry Unit 6

Rapid Revision

WCH16 — Practical Skills in Chemistry II

Last-minute cram sheets — designed to be scanned, not read

Condensed from the full A* guide (20 past papers, Jun 2019 - Jun 2026)

50 marks · 1 h 20 min · 3–5 questions · about **1.5 minutes per mark**

Bring a ruler

Unit 6 repeats the same tests, techniques and calculation chains every series — it is the most learnable of the three A2 units.

TRIAGE — WHAT IS ON THE PAPER

Frequency	Content
Almost every paper	Inorganic ion / transition-metal identification · organic functional-group tests (often with NMR / MS / IR) · a titration calculation · an organic preparation (reflux → separate → dry → distil or recrystallise → melting point → % yield)
Most papers	A kinetics practical with a graph — clock reaction, sampling and titration, half-lives, Arrhenius
Sometimes	Enthalpy of solution · electrochemical cell · combustion analysis · thermal decomposition · water of crystallisation

Priority order: (1) cation/anion tests and transition-metal colours · (2) organic tests and distinguishing pairs · (3) titration technique and calculation chains · (4) recrystallisation, filtration, reflux, distillation, separating, drying, melting point · (5) kinetics explanations and graphs · (6) hazards.

THE SIX RULES OF OBSERVATION ANSWERS

1. Say "**precipitate**" for every solid that forms, **with its colour**. Never call a solution a precipitate — "**orange solution**", not "orange ppt".
2. Give **both the before and after** colour: "**orange → green**".
3. Use the scoring colour words: "**(pale) pink**" for the KMnO_4 end point (**purple rejected**) · "**dark / deep blue**" for copper-ammonia (just "blue" rejected) · "**colourless**" (never "clear") · "**pale yellow ppt**" for iodoform (not "bright yellow").
4. **Fumes and smoke:** HCl is **misty / steamy fumes**; $\text{NH}_3 + \text{HCl}$ gives **white smoke**; NO_2 is **brown fumes** (not "brown smoke"). **Don't swap them.**
5. **Negative results:** "**no change / stays blue / stays orange**" — "none" or "no result" is penalised.
6. **The list principle:** **an extra wrong observation cancels a right one.** Write only what you are sure of.

SHEET 1 — GAS, ANION & CATION TESTS

Gas	Test	Result
NH ₃	Damp red litmus held in the vapour (not dipped in)	Turns blue ; or white smoke with conc. HCl on a glass rod
CO ₂	Limewater	Cloudy / milky
O ₂ / H ₂	Glowing / lighted splint	Relights / squeaky pop
HCl	Damp blue litmus; NH ₃	Red; white smoke (NH ₄ Cl)
Water vapour	Cobalt chloride paper	blue → pink (reject red) ; condensation at the mouth of the tube

Name the gas as NH₃ and the ion as NH₄⁺ — the schemes reject swapping them ("ammonia ion" rejected).

Ion	Test	Positive result
CO ₃ ²⁻	Dilute acid	Effervescence; limewater milky
SO ₄ ²⁻	Dilute HCl (or HNO ₃) then BaCl ₂ (aq)	White precipitate
Cl ⁻ / Br ⁻ / I ⁻	Dilute HNO₃ then AgNO ₃ (aq)	White (dissolves in dilute NH ₃) / cream (conc. NH ₃ only) / yellow (insoluble even in conc. NH ₃)
NO ₃ ⁻	NaOH + aluminium, warm	NH ₃ — red litmus turns blue
CH ₃ COO ⁻	Warm with conc. H ₂ SO ₄ , then add ethanol	Vinegar smell → fruity smell (ester)

WHY ACIDIFY FIRST — AND THE TWO ZERO-SCORING ERRORS

The acid **removes other ions that would also precipitate** — CO₃²⁻, HCO₃⁻, SO₃²⁻ (with Ba²⁺), or OH⁻/CO₃²⁻ (with Ag⁺).
NEVER acidify a sulfate test with H₂SO₄ — it adds sulfate, so the test is always positive. **NEVER use HCl before AgNO₃**.
AgCl and AgBr "**colours are hard to distinguish**" (not "the same") → confirm with dilute NH₃. If another precipitate is present, **separate the silver halide first**. Conc. H₂SO₄ + a chloride → **steamy fumes of HCl**, not white smoke.

Flame tests: Li⁺ red · Na⁺ yellow · K⁺ lilac · Ca²⁺ orange-red (brick red) · Sr²⁺ red · Ba²⁺ green · Cu²⁺ blue-green. Write the ion, not the metal.

Ion (colour)	Few drops NaOH / NH ₃	Excess NaOH	Excess NH ₃
Cu ²⁺ (blue)	pale blue ppt	no change	dissolves → deep/dark blue [Cu(NH ₃) ₄ (H ₂ O) ₂] ²⁺
Fe ²⁺ (pale green)	green ppt, turns brown on standing (air oxidation)	no change	no change
Fe ³⁺ (yellow-brown)	(red-/orange-)brown ppt	no change	no change
Cr ³⁺ (green)	grey-green ppt	dissolves → dark green [Cr(OH) ₆] ³⁻	dissolves → violet/purple [Cr(NH ₃) ₆] ³⁺
Zn ²⁺ (colourless)	white ppt	dissolves → colourless [Zn(OH) ₄] ²⁻	dissolves → colourless [Zn(NH ₃) ₄] ²⁺
Mn ²⁺ (pale pink)	off-white/pale brown ppt, darkens in air	no change	no change
Co ²⁺ (pink)	blue ppt, then pink on standing	no change	dissolves → yellow/straw, darkens in air

REACTION TYPES — A FAVOURITE QUESTION

Precipitate forming with OH⁻ or a little NH₃ = **DEPROTONATION (acid-base)**, not ligand exchange · **dissolving in excess NaOH** (Cr³⁺, Zn²⁺) = deprotonation / **amphoteric** · **dissolving in excess NH₃** (Cu²⁺, Cr³⁺, Zn²⁺, Co²⁺, Ni²⁺) = **LIGAND EXCHANGE** · **darkening/browning in air** = **oxidation**.

SHEET 2 — TRANSITION-METAL CHEMISTRY EXAMINED

Copper: Cu^{2+} + excess conc. HCl → blue → green → **yellow** $[\text{CuCl}_4]^{2-}$ (ligand exchange) · $\text{Cu}(\text{OH})_2$ heated → **black** CuO · **never write** $[\text{Cu}(\text{NH}_3)_6]^{2+}$ — **always rejected** · Fehling's red precipitate = **copper(I) oxide, Cu_2O** · $\text{Cu}^{2+} + \text{I}^- \rightarrow \text{CuI}$ (white ppt) + I_2 .
Chromium: Cr^{3+} (green) + excess $\text{NaOH} \rightarrow [\text{Cr}(\text{OH})_6]^{3-}$ (dark green); + **H_2O_2 warm** → CrO_4^{2-} **yellow** (H_2O_2 is the oxidising agent, +3 → +6); + acid → $\text{Cr}_2\text{O}_7^{2-}$ **orange**. $\text{Cr}_2\text{O}_7^{2-}$ + ethanol/warm → Cr^{3+} green. **The chromate/dichromate interconversion is acid-base, not redox.** Yellow **BaCrO_4** from $\text{CrO}_4^{2-} + \text{Ba}^{2+}$.
Cobalt: pink $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ + conc. $\text{HCl} \rightarrow$ **blue** $[\text{CoCl}_4]^{2-}$. **Iron:** Fe^{2+} + oxidant → green → **yellow/yellow-brown** (reject red); Fe^{3+} oxidises I^- to I_2 . **Manganese:** MnO_4^- purple; Mn^{2+} very pale pink; **MnO_2 brown — forms if there is too little acid in a KMnO_4 titration.**

Vanadium state	Ion	Colour
+5	VO_2^+	yellow
+4	VO^{2+}	blue
+3	V^{3+}	green
+2	V^{2+}	violet / purple (reject lilac)

Reduce with **zinc in acid**; heat to speed it up. **The first green is a mixture of yellow VO_2^+ and blue VO^{2+} , not V^{3+} .** V^{2+} left in air turns green — oxidised to V^{3+} by O_2 , which is not a strong enough oxidant to go further.

Water of crystallisation (4 marks): weigh the sample → heat in a **crucible or evaporating basin (not a beaker or test tube)** → **heat to constant mass** → mass of water = difference. $x = \text{mol water} \div \text{mol anhydrous salt}$. **Observations on heating a hydrate:** condensation at the mouth of the tube; cobalt chloride paper **blue → pink**; the solid dissolves in its own water of crystallisation. **Reject "effervescence".**

Heating a nitrate: brown fumes (NO_2): O_2 relights a glowing splint; nitrates are **oxidising** — a splint in contact with the solid can make it burn uncontrollably.

SHEET 3 — ORGANIC TESTS & DISTINGUISHING PAIRS

Test	Positive result (exact wording)	Shows
Bromine water	orange/brown → colourless (reject "clear")	C=C
Acidified KMnO ₄	purple → colourless (reject violet)	C=C
Burn in air	Very smoky (sooty) flame	Benzene ring (high C:H)
Sodium	Effervescence (H ₂)	–OH (alcohol or acid)
PCl ₅	Misty / steamy fumes of HCl	–OH
Na₂CO₃ / NaHCO₃ (named in full)	Effervescence; gas turns limewater milky	–COOH
Acidified K ₂ Cr ₂ O ₇ , warm	orange → green	1° or 2° alcohol, or aldehyde
2,4-DNPH (Brady's)	yellow/orange/red ppt (reject brick red, brown)	C=O — aldehyde or ketone
Tollens', warm	Silver mirror / grey-black solid (not just "mirror")	Aldehyde
Fehling's / Benedict's, warm	blue → (brick-) red precipitate (Cu ₂ O)	Aldehyde
I₂ + NaOH , warm	Pale yellow precipitate (CHI ₃), antiseptic smell	CH₃CO- or CH₃CH(OH)-
Carboxylic acid + conc. H ₂ SO ₄ , warm	Sweet / fruity smell	Alcohol (an ester forms)
Water, then conc. NH ₃ on a glass rod	Misty fumes, then white smoke	Acyl chloride
CuSO ₄ (aq) + amine	pale blue ppt → deep blue solution	Amine

THE DEDUCTIONS THE SCHEMES REWARD

DNPH positive + **dichromate no change** (or Fehling's stays blue) → **ketone**; + **iodoform positive** → **methyl ketone CH₃CO-**.
Na fizzes but dichromate stays orange → **tertiary alcohol**.
PCl₅ fumes but no NaHCO₃ fizz → **alcohol, not acid**.
Iodoform positive but not a ketone → **CH₃CH(OH)- secondary alcohol, or ethanal**.
Rejected: "hydroxide / OH-" for the hydroxyl group · "benzene / phenyl" for an alkene · "**carbonyl**" alone when "**aldehyde or ketone**" is needed.
Detecting an ester in an acid mixture: add NaHCO₃ to **neutralise the acid, whose smell masks the fruity ester smell**.

Distinguishing pairs — give the reagent AND both results. The observation mark **depends on the right reagent**. **Na and PCl₅ cannot separate an acid from an alcohol — both react.**

aldehyde vs ketone → **Tollens'/Fehling's/dichromate:** mirror / red ppt / orange→green vs **no change** · methyl ketone vs other ketone → **I₂ + NaOH:** pale yellow ppt vs no change · acid vs alcohol → **Na₂CO₃:** effervescence vs none · Fe²⁺ vs Cr³⁺ → **NaOH then excess:** both green ppt, but **Cr(OH)₃ dissolves in excess and Fe(OH)₂ doesn't.**

Identifying a carbonyl by its derivative: react with 2,4-DNPH → **recrystallise** the precipitate → **measure its melting temperature** → compare with data-book values.

Spectra: fragments 15 CH₃⁺ · 29 C₂H₅⁺/CHO⁺ · 43 CH₃CO⁺ · 45 COOH⁺ · 59 COOCH₃⁺ · 77 C₆H₅⁺. **IR:** alcohol O–H 3750–3200 · **acid O–H 3300–2500 broad** · N–H 3500–3300 · **ester C=O 1750–1735** · acid C=O 1725–1700 · amide C=O 1700–1630. **¹H patterns:** triplet(3H)+quartet(2H) = CH₃CH₂– · doublet(6H)+septet(1H) = (CH₃)₂CH– · **9H singlet = (CH₃)₃C–** · singlet ~2.1–2.4 = CH₃CO– · ~9–10 = CHO · ~10–12 = COOH.

Combustion analysis: mass C = m(CO₂) × 12/44; mass H = m(H₂O) × 2/18; O by difference. **Water is absorbed by anhydrous CaCl₂, CO₂ by soda lime** (not limewater).

SHEET 4 — TITRATION & QUANTITATIVE WORK

Need	Use	Reject
Accurately known volume (aliquot)	Volumetric pipette + filler	Measuring cylinder; burette for sampling
Variable / non-standard volumes (7.5, 11.0 cm ³)	Burette or graduated pipette	"Just pipette"; volumetric pipette
Excess / approximate volume	Measuring cylinder — quicker, precision not needed	—
Adding dropwise	Teat (dropping) pipette	"Small amounts"

Making a standard solution (3 marks): dissolve the weighed solid in distilled water **in a beaker** → transfer **the solution and the washings** to a volumetric flask → make up to the mark → **stopper and invert** to mix.

Accurate end point (any 3): add **dropwise near the end point** · swirl · **white tile** · **rinse the flask walls / burette tip** with distilled water · read the **bottom of the meniscus at eye level** · **rinse the burette with the titrant** so it isn't diluted.

Reading the burette from the top of the liquid: the titre is **unchanged** — it is the **difference between two readings taken the same way**.

Titration	Indicator and end point
Strong acid → alkali	Methyl orange: yellow → orange (reject yellow → red)
Weak acid ← NaOH	Phenolphthalein: colourless → pink (methyl orange / litmus rejected)
KMnO₄	Self-indicating: colourless → first permanent PALE PINK (reject purple)
Thiosulfate / iodine	Starch, added when PALE YELLOW / STRAW (not "yellow", not "brown"); blue-black → colourless

THE KMnO₄ RULES

Acidify with dilute H₂SO₄ ONLY — HCl is oxidised to Cl₂ (titre too big); HNO₃ oxidises Fe²⁺. Too little acid → brown MnO₂. KMnO₄ goes in the burette.

KMnO₄ + ethanedioate is warmed (~60 °C) because of a **high activation energy**; then the **Mn²⁺ produced autocatalyses** it.

Why starch is added late: at high [I₂] it forms an **insoluble complex**, making the end point hard to see.

Mean titre: every burette reading to **2 d.p. ending in 0 or 5**. Mean of **concordant titres only** (within 0.10–0.20 cm³); **exclude the rough and any outlier**.

Keep 2 d.p. — 12.65, not 12.7.

% uncertainty = (uncertainty × number of readings ÷ value) × 100. Burette titre: **2 × 0.05 ÷ titre × 100** — **forgetting the ×2 is rejected**. Temperature change: **2 × 0.25 ÷ ΔT**. Pipette: **0.06 ÷ 10.0 (one reading)**.

The calculation chain: moles of titrant → **ratio** → moles in the aliquot → **× scale-up (×10 for 25.0 cm³ of 250 cm³)** → mass / concentration / % → 2–3 s.f. with units. **The scale-up is the most-lost mark.** **Back titration:** moles added – moles left (scaled) = moles reacted → ÷ ratio.

ERRORS — ALWAYS STATE THE DIRECTION AND THE KNOCK-ON

NH₃ not fully boiled off → **reacts with the HCl** → **titre larger** → **calculated % lower**.

HNO₃ not neutralised in a brass analysis → **oxidises I⁻ to extra I₂** → **larger titre** → **higher % Cu**.

Cl⁻ not removed before KMnO₄ → **MnO₄⁻ oxidises Cl⁻** → **larger titre** → **lower calculated KClO₃**.

Precipitate not washed or not dry → **mass too high** → **moles too high**.

"You must show your working" → **a correct answer with no working scores 0**.

Enthalpy: polystyrene cup = **good insulator** → **less heat loss (reject "low specific heat capacity"; "prevents heat loss" ignored)**. $q = mc\Delta T$ with **m = mass of solution** (not + solid); **ΔH = -q ÷ n**, with sign and kJ mol⁻¹. **Cooling correction (5 marks):** record every 30 s for ~3 min **before** mixing → add at a set time → record every 30 s for ~5 min more → plot temperature against time → **extrapolate both best-fit lines to the time of mixing** → read ΔT.

Cells: each metal in **1.00 mol dm⁻³ of its own ion** · salt bridge = **filter paper soaked in saturated KNO₃** · **high-resistance voltmeter (reject "voltmeter", battery, ammeter)** · 298 K · **platinum** for an ion/ion couple. **KOH or KCl can be wrong** — OH⁻ precipitates metal ions, Cl⁻ reacts with Ag⁺.

Other: mass = volume × density (not M_r ÷ density) · mol = V ÷ 24 000 cm³ at r.t.p. · **a saturated solution needs time to reach equilibrium** · **K_a from a pH curve: read the pH at half-equivalence (show it on the graph) → K_a = 10^{-pH}**.

SHEET 5 — KINETICS PRACTICALS & GRAPHS

CLOCK REACTIONS — THE EXPLANATIONS, WITH THEIR REJECTIONS

Thiosulfate (or phenol) **reacts with the I_2 (or Br_2) as it forms**, so the colour appears only **after a fixed amount of reaction**, when it is used up. **Reject "slows the reaction" / "reacts with I^- ".**

The amount is **small**, so **reactant concentrations barely change** before the colour appears → **$1/t$ measures the initial rate**.

Keep the total volume constant (add water) so the **volume of a reagent is proportional to its concentration**. Keep **temperature** constant (it changes k).

Colour change: **colourless → blue-black**; view against a **white background**. Mix quickly (a burette delivers too slowly); pour back and forth.

Other controls: other reactants **in large excess** → their concentrations stay **effectively constant**, so the rate depends only on the one studied (**just "in excess" is ignored**); the order found is then **only for that reactant, not the overall order**.

Quenching: $NaHCO_3$ **neutralises the H^+ catalyst**; ice slows it; excess KI removes MnO_4^- instantly. **NaOH cannot be used for acid-catalysed iodination — alkali also catalyses it.**

Ethanol as a co-solvent dissolves both the halogenoalkane and NaOH so they mix — **it is not a catalyst**. **Titre $\propto$ [reactant]**, so titre can be plotted instead of concentration.

GRAPHS — USUALLY 3 MARKS

1. Axes the right way round, **labelled with quantity / unit (t / s, not T / s)**, scale using $\geq$ **half the grid in both directions**.

2. Points within $\pm\frac{1}{2}$ -1 small square.

3. The right line: **straight best-fit line / straight line through the origin / smooth curve — never point-to-point; ignore the anomalous point**.

Graph	Conclusion
[X] or titre vs time, straight line	Zero order in X
$1/t$ (rate) vs volume or [X], straight line through the origin	First order in X
conc vs time with constant successive half-lives (drawn on the graph)	First order
Slow → fast → slow (sigmoid)	Autocatalysis — a product (Mn^{2+}) is the catalyst; slows as reactants are used up

Rate at time t : draw a **tangent** and find its gradient, **with units** ($cm^3\ h^{-1}$). **Half-lives in the units of the axis** (s, not min).

Activation energy: plot **$\ln k$ (or $\ln(1/t)$) against $1/T$** , T in K (reject $\ln(1/T)$) → best-fit line → **gradient = $-E_a/R$ → $E_a = -\text{gradient} \times R$, positive**, in $kJ\ mol^{-1}$ to 2-3 s.f. **In A is the y-intercept at $1/T = 0$ — not where the plotted axis happens to start.**

SHEET 6 — PREPARATION, PURIFICATION & HAZARDS

Reflux — why (2 marks): heating increases the rate / lets the reaction go to completion; the **vertical condenser condenses vapour and returns it**, so **volatile reactants and products are not lost**.

Diagram (3 marks): round-bottomed or pear-shaped flask with liquid and heat · **vertical condenser, water in at the bottom, out at the top** · **open at the top, no gaps**.

Classic errors to spot: condenser sealed at the top (pressure builds) · water flow reversed · conical flask · no anti-bumping granules · thermometer in the condenser.

Heat source: a **water bath or electric heater** for flammable liquids — **never a Bunsen**. **Anti-bumping granules: smooth boiling / prevent superheating** ("prevent bumping" alone is ignored; **reject "explosion"**). **Stirring** is needed when reactants are **immiscible**.

Distillation errors: conical flask → pear-shaped/round · **thermometer bulb level with the condenser inlet**, not in the liquid · sealed collecting vessel → **open or vented** · water flow reversed. **Collect a range that BRACKETS the boiling point** (186–190 °C for b.p. 188) — a single value, or a range **ending** at 188, is rejected. **Boiling point varies on different days because atmospheric pressure varies**.

Steam distillation is used because the compound **decomposes at its boiling point**; with steam it **distils below** it. A **vertical open tube** prevents pressure build-up. The distillate is **cloudy / two layers** because the product is **only partly miscible** with water.

SEPARATING FUNNEL AND DRYING

Washing with $\text{NaHCO}_3/\text{Na}_2\text{CO}_3$ neutralises and removes the acid. **Stopper, invert, shake, then open the tap to release pressure from CO_2** . **Remove the stopper before draining**, otherwise air can't enter and the liquid won't flow.

Salting out: adding NaCl **lowers the organic product's solubility in water**, helping the layers separate — **it is not a drying agent**.

Drying liquids: anhydrous MgSO_4 , Na_2SO_4 , CaCl_2 or CaSO_4 (KOH pellets for amines). **Reject CuSO_4 , CoCl_2 , conc. H_2SO_4 , CaO** . Enough when the liquid is **clear (no longer cloudy)** and the agent **stays free-flowing**. Then decant/filter and distil.

Drying solids: desiccator, **warm (low-temperature) oven**, or between filter papers — **not a hot oven** (decomposition/melting) and **not a solid drying agent** (it can't be separated).

RECRYSTALLISATION — THE ORDER IS THE MARK

1. Dissolve in the **minimum volume of HOT solvent** — so the solution is **saturated** and as little product as possible stays dissolved on cooling.

2. Filter while HOT (preheated funnel) — removes **INSOLUBLE** impurities.

3. Cool (ice) so the product crystallises.

4. Filter under reduced pressure — **SOLUBLE** impurities stay in the filtrate.

5. Wash with a small volume of ICE-COLD solvent and dry.

Hot filtration = insoluble; cold filtration = soluble. Reversing them scores 0, and "removes impurities" alone scores 0. Never "evaporate off the solvent".

Büchner diagram (3 marks): funnel with a **perforated base and FLAT filter paper covering all the holes (reject fluted paper)** · flask with a **side arm**, sealed with a bung · side arm to a **vacuum pump**. **Advantages: faster and a drier product** — **reject "removes more impurities"**.

Melting point: dry sample in a **capillary tube sealed at one end**, next to the thermometer bulb in melting-point apparatus or a **Thiele tube of oil** — a **water bath is useless above 100 °C**. Heat **quickly to ~10–20 °C below**, then slowly. **Pure: sharp and matches the data book. Impure: LOWER AND over a WIDER range** — **both points are needed. Never boiling point as a purity test**.

Yield: mass of a liquid = **volume × density** → moles of each reactant → **limiting reagent** → theoretical mass → % yield (2–3 s.f.). **>100% means the product is still damp** — not "hydrated salt". Reasons for <100%: **equilibrium** (esters) · side reactions · incomplete reaction · **loss on transfer (say where)** · product left dissolved in the recrystallisation solvent.

Hazard	Examples	Precaution
Corrosive	conc. $\text{HCl}/\text{HNO}_3/\text{H}_2\text{SO}_4$, NaOH , phenol, phenylamine, NH_3	Gloves
Toxic by inhalation	SO_2 , NO_2 , NH_3 , HCl fumes, Br_2 , acyl chlorides	Fume cupboard
Toxic by skin absorption	nitrobenzene, phenol	Gloves
Flammable	ether, ethanol, hexane	No naked flames; water bath / electric heater
Oxidising	conc. HNO_3 , metal nitrates, KMnO_4	Keep away from combustible materials
Harmful to the environment	organic waste, phenylamine	Collect as waste — not down the sink

The precaution must match the named hazard of a named substance. Goggles and lab coats are assumed and score nothing. Name hazards exactly: "oxidising" (**reject "oxidative", "oxidised"**) · "corrosive" for conc. acids (**reject "flammable"**). **Mismatches score 0:** "toxic, so wear gloves" unless it is toxic by skin absorption.

Explanations: cold water / slow addition **because the reaction is exothermic (reject "explosive")** · generate SO_2 **in situ** so only a small controlled amount is made · use the **solid salt** rather than a toxic liquid.

SHEET 7 — THE 25 MOST LOST MARKS

1. Not writing "**precipitate**"; calling a solution a precipitate.
2. **Purple** for the KMnO_4 end point — it is **pale pink**.
3. "Blue" instead of **deep/dark blue** for copper-ammonia; writing $[\text{Cu}(\text{NH}_3)_6]^{2+}$.
4. "Yellow" instead of **pale yellow / straw** for when to add starch.
5. "**Clear**" instead of colourless · "bright yellow" for iodoform · "brick red" for DNPH · "lilac" for V^{2+} · "red" for cobalt chloride paper.
5. Swapping **misty fumes (HCl)** and **white smoke (NH_4Cl)**; writing "brown smoke".
7. **H_2SO_4 to acidify a sulfate test; HCl before AgNO_3** or in a KMnO_4 titration.
3. "Dissolves" **without saying in excess**.
3. **Ligand exchange for precipitate formation** — it is **deprotonation**.
2. Averaging **non-concordant** titres; mean not to **2 d.p.**
1. Forgetting the **$\times 10$ scale-up**, the reaction ratio, or **$[\text{OH}^-] = 2c$** for $\text{Ca}(\text{OH})_2$.
2. Forgetting **$\times 2$** in a burette or temperature-change uncertainty.
3. **Mass = $M_r \times$ density** instead of **volume $\times$ density**.
1. **Conical flask** in reflux or distillation; **sealed apparatus**; water flow reversed; thermometer bulb in the liquid.
5. **Fluted filter paper** or **no side arm** in a Büchner diagram.
3. **Hot and cold filtration reversed**, or "removes impurities" **without soluble / insoluble**.
7. Saying washing removes "**insoluble**" impurities — it removes **soluble** ones.
3. **Boiling point** as a purity test; only "lower" **or** only "range" for an impure melting point.
3. **CuSO_4 or conc. H_2SO_4** as a drying agent; a **hot oven** for crystals.
2. "Prevent bumping", "to keep cool", "increase rate", "explosive" — **vague reasons that are ignored**.
1. **H_2SO_4 as "initiator" or "provides H^+ "** — it is a **catalyst**.
2. "In excess" **without** "so its concentration stays constant / the rate depends only on...".
3. **Negative E_a** ; half-lives in the wrong units; **no construction lines on the graph**.
4. Hazard and precaution **mismatched**; "oxidative"; generic "wear goggles".
5. **Measuring cylinder for an accurate aliquot** (or a pipette for an excess reagent).

THE ROUTINE FOR EVERY QUESTION

1. **Observation:** reagent → what you **SEE** (precipitate + colour / solution colour / fumes / effervescence), **before AND after**.
2. **Deduction:** name the ion or functional group, **with its charge**.
3. **Technique:** name the apparatus, say **HOW** (dropwise, to the mark, invert, vent) and **WHY** (soluble vs insoluble, exothermic, volatile).
4. **Calculation:** moles → ratio → **scale-up ($\times 10$)** → answer → **units** → s.f. Mean titre from **concordant titres only**.
5. **Graph:** axes + units + half the grid; points; correct line; **show construction lines**.
6. **Safety:** named substance + **its** hazard + a **matching** precaution.

Pace: 50 marks in 80 minutes — about 1.5 minutes per mark. Draw apparatus **with a ruler, no gaps, labelled, and make sure it would work** — open to air when heating, water in at the bottom.